Scratch Cat FX

Studio 87. Free and open source (AGPL-3.0-or-later).

Intro

Scratch Cat FX is a scratch effect plugin. It records the sound coming from your DAW and lets you scratch, brake and backspin it like a record, then adds EQ, colour, PurrOne style Drive, Transistor and Fuzz, and a 3 slot FX chain. When you do not touch it, the sound passes straight through unchanged.

Quick start

- Put the plugin on a track or the master and play your project.
- Turn on Hand on platter, then move Platter speed (or automate or MIDI-map it in your host) to scratch the last part of the sound. Turn Hand on platter off to let it play on at normal speed.
- Brake slows the sound to a stop, Backspin spins it backward and stops, Go live returns to the live DAW sound.
- Note: this version uses the host's plain parameter view as its window. The designed window (platter, hot cues, mixer) is shown in the mockup in the repository and is not built into the plugin yet.

Scratch controls

- Hand on platter: while on, the platter follows Platter speed instead of playing normally.
- Platter speed: -4 to +4 times normal speed. Negative plays backward.
- Brake, Backspin, Go live: described above. Go live returns to the live sound.

Mixer and colour

- Trim: input level, -12 to +12 dB.
- Low, Mid, High: three band EQ, -26 to +12 dB. All the way down kills the band.
- Colour: left is a low pass filter, right is a high pass filter, centre is off.
- Master: output level, -24 to +12 dB.

PurrOne colour

- PurrOne colour on switches the stage on. Drive is a symmetric analog style clip, Transistor is an asymmetric stage with even harmonics, Fuzz is a gritty octave-rich fuzz. PurrOne mix blends it with the clean sound.

FX chain

- Three slots in series, each with a Type (Off, Filter, Echo, Reverb, Flanger, Crush), an On switch, an Amount and a Mix. FX chain mix scales all three. Echo beats sets the echo time in beats.

Recommendations

- Start with the chain mix near 60 percent and add one effect at a time.
- Use Brake and Backspin for build-up endings, and automate Hand on platter and Platter speed for scratches.
- Nothing here has been tuned by ear yet: please report anything that sounds wrong on GitHub.

Interface tour

Every pink number marks a control that has an info bubble in the interactive mockup (docs/mockup), and the tables below carry the text of those bubbles. The pictures come from the mockup, which mirrors the plugin window; the compiled plugin can differ by a few pixels of layout.



Header bar

1	**Previous preset**	Previous preset.
2	**Init**	Open the preset list: factory presets and your own.
3	**Next preset**	Next preset.
4	**Save**	Save the current settings as your own preset (stays on this device).
5	**Delete, export or import your presets**	Delete, export or import your presets.
6	**Guide**	Quick guide: what each part does.
7	**Settings**	Info bubbles and eco mode.
8	**About & Support**	About Studio 87, privacy, and ways to support the project.

Window

9	**Source**	What the plugin hears from your DAW. In a DAW this is the track you put Scratch Cat FX on.
10	**DAW play**	Start or stop the track as if your DAW were playing it. Shortcut: P.
11	**Load file**	Load your own audio file (wav, mp3, flac, ogg). It stays in this browser tab and is never uploaded.
12	**Capture**	Grab the last bars of the DAW input into the scratch deck. Here it grabs the whole test loop.
13	**Host BPM**	The tempo your DAW runs at. Sync matches the deck to it. Double-click resets.
14	**Waveform**	Grab the waveform and drag it like a record to scratch. Beat lines are every beat; taller lines are bars. Mouse wheel nudges.
15	**2 BARS**	Show 2 bars across the waveform.
16	**4 BARS**	Show 4 bars across the waveform.
17	**8 BARS**	Show 8 bars across the waveform.
18	**16 BARS**	Show 16 bars across the waveform.
19	**Delete**	Delete switch: while on, clicking a set hot cue removes it. Holding Shift while clicking does the same.

20	**Hot cue 1**	Hot cue 1. Empty: click to set a marker here. Set: click to jump to it. Shift-click, or the Delete switch, removes it. Right-click edits name, colour, position and on/off.
21	**Hot cue 2**	Hot cue 2. Empty: click to set a marker here. Set: click to jump to it. Shift-click, or the Delete switch, removes it. Right-click edits name, colour, position and on/off.
22	**Hot cue 3**	Hot cue 3. Empty: click to set a marker here. Set: click to jump to it. Shift-click, or the Delete switch, removes it. Right-click edits name, colour, position and on/off.
23	**Hot cue 4**	Hot cue 4. Empty: click to set a marker here. Set: click to jump to it. Shift-click, or the Delete switch, removes it. Right-click edits name, colour, position and on/off.
24	**Hot cue 5**	Hot cue 5. Empty: click to set a marker here. Set: click to jump to it. Shift-click, or the Delete switch, removes it. Right-click edits name, colour, position and on/off.
25	**Hot cue 6**	Hot cue 6. Empty: click to set a marker here. Set: click to jump to it. Shift-click, or the Delete switch, removes it. Right-click edits name, colour, position and on/off.
26	**Hot cue 7**	Hot cue 7. Empty: click to set a marker here. Set: click to jump to it. Shift-click, or the Delete switch, removes it. Right-click edits name, colour, position and on/off.
27	**Hot cue 8**	Hot cue 8. Empty: click to set a marker here. Set: click to jump to it. Shift-click, or the Delete switch, removes it. Right-click edits name, colour, position and on/off.
28	**Vinyl**	On: pausing slows the record down like a turntable. Off: it stops at once.
29	**Trans.**	Transform gate: chops the sound on and off in time with the beat.
30	**Stutter**	Hold: retriggers the cue point every 1/16 beat.
31	**Brake**	Slows the record to a stop.
32	**Backspin**	The record spins back fast, then stops.
33	**Roll**	Hold: rolls a tiny loop of 1/8 beat; letting go lands where the song would have been.
34	**Censor**	Hold: plays backwards; letting go jumps to where the song would have been.
35	**Loop of 1/4 beat from here**	Loop of 1/4 beat from here. Press again to leave the loop.
36	**Loop of 1/2 beat from here**	Loop of 1/2 beat from here. Press again to leave the loop.
37	**Loop of 1 beat from here**	Loop of 1 beat from here. Press again to leave the loop.
38	**Loop of 2 beats from here**	Loop of 2 beats from here. Press again to leave the loop.
39	**Loop of 4 beats from here**	Loop of 4 beats from here. Press again to leave the loop.
40	**Loop of 8 beats from here**	Loop of 8 beats from here. Press again to leave the loop.
41	**Loop in**	Set the loop start here.
42	**Loop out**	Set the loop end here and start looping.
43	**Exit**	Leave the loop.
44	**CUE**	CUE. While playing: jumps back to the cue point and pauses. While paused: sets the cue point here; hold to preview from it.
45	**Play or pause the deck**	Play or pause the deck.
46	**Sync**	Match this deck's tempo to the host BPM (or to the master deck) and keep it matched.
47	**Key lock**	Key lock: change the tempo without changing the pitch.
48	**Quant**	Quantize: cue points, hot cues and loops snap to the nearest beat.
49	**Tempo**	Speed of this deck. Up is faster. The range buttons set how far it goes. Double-click resets.
50	**How far the tempo fader goes (percent)**	How far the tempo fader goes (percent).
51	**16**	How far the tempo fader goes (percent).
52	**50**	How far the tempo fader goes (percent).
53	**Reset**	Put the tempo fader back to zero.

78	**Window 100 %**	Window size. Drag any edge or corner of the plugin to resize it; the shape stays the same. Double-click here for 100 %.
79	**Resize**	Drag to resize the window. Every edge and corner works, and the shape stays the same.

Mixer

54	**Trim**	Input gain of the deck. Double-click resets.
55	**High**	High shelf (4 kHz). Double-click resets.
56	**Mid**	Mid band (1 kHz). Double-click resets.
57	**Low**	Low shelf (250 Hz). Double-click resets.
58	**Color**	One-knob filter: left is low-pass, right is high-pass. Double-click resets.
59	**High**	Silences the highs. Keep it pressed during a drop swap.
60	**Mid**	Silences the mids.
61	**Low**	Silences the lows.
62	**Deck**	Channel fader of the scratch deck. Double-click resets.
63	**Master**	Output level. Double-click resets.
64	**Cross**	Crossfader: left is the scratch deck, right is the DAW input playing straight. Double-click resets.
65	**Curve**	Smooth: both full in the middle. Sharp: cuts like a battle mixer. Constant: equal power.
66	**FX on**	Send the effects to the scratch deck only, or to everything.
67	**FX 1**	Turn this effect slot on or off.
68	**Type**	What this slot does.
69	**Amount**	Filter: sweep from low-pass to high-pass. Echo: delay time in beats. Reverb: size. Flanger: rate. Crush: amount. Double-click resets.
70	**Dry/Wet**	How much of the effect is mixed in. Double-click resets.
71	**FX 2**	Turn this effect slot on or off.
72	**FX 3**	Turn this effect slot on or off.
73	**On**	Turns the PurrOne colour stage on or off. It sits after the three effects and before the master.
74	**Drive**	Smooth tanh saturation. Keep it under 80 % for warm, over 80 % for hard. Double-click resets.
75	**Transistor**	Lopsided transistor curve: adds even harmonics, a thicker, warmer edge. Double-click resets.
76	**Fuzz**	Fuzz: biased clipping with an octave-up growl. Nasty above 80 %. Double-click resets.
77	**Mix**	Dry/wet of the colour stage. Double-click resets.